

Nonbinary LDPC Coding for Multicarrier Underwater Acoustic Communication

Jie Huang, Shengli Zhou, *Member, IEEE*, and Peter Willett, *Fellow, IEEE*

Abstract—Recently, multicarrier modulation in the form of orthogonal frequency division multiplexing (OFDM) has been shown feasible for underwater acoustic communications via effective algorithms to handle the channel time-variability. In this paper, we propose to use nonbinary low density parity check (LDPC) codes to address two other main issues in OFDM: (i) plain (or uncoded) OFDM has poor performance in fading channels, and (ii) OFDM transmission has high peak to average power ratio (PAPR). We develop new methods to construct nonbinary regular and irregular LDPC codes that achieve excellent performance, match well with the underlying modulation, and can be encoded in linear time and in a parallel fashion. Based on the fact that the generator matrix of LDPC codes has high density, we further show how to reduce the PAPR considerably with minimal overhead. Experimental results confirm the excellent performance of the proposed nonbinary LDPC codes in multicarrier underwater acoustic communications.

Index Terms—Channel coding, LDPC, regular cycle code, multicarrier, underwater acoustic communication

I. INTRODUCTION

MULTICARRIER underwater acoustic communication, in the form of orthogonal frequency division multiplexing (OFDM), has been actively investigated recently [3]–[8]. This is motivated by its unique capability to handle high-rate transmissions over long dispersive channels. Specifically, OFDM divides the available bandwidth into a large number of overlapping subbands, so that the symbol duration is long compared to the multipath spread of the channel. As a result, inter-symbol-interference (ISI) may be neglected in each subband, which greatly reduces the complexity of channel equalization at the receiver.

However, to make OFDM modulation successful in a practical underwater system, the following three issues must be adequately addressed.

- 1) OFDM is sensitive to intercarrier interference (ICI) caused by channel variations. Underwater channels vary fast due to the large ratio of the platform motion relative to the sound propagation speed. Even with stationary transmitters and receivers, significant ICI could still exist due to wave action and water motion.

- 2) Plain (or uncoded) OFDM has poor performance in fading channels, since it does not exploit the multipath diversity inherent to the channel.
- 3) OFDM transmission has large peak-to-average power ratio (PAPR). The required large power backoff reduces the transmission range.

In [7], we have developed a two-step approach that can effectively deal with fast-varying underwater channels with nonuniform Doppler shifts. Experiment results in [7] have demonstrated that OFDM is one viable solution for high rate transmissions over time-varying underwater acoustic channels.

In this paper, we investigate the performance and PAPR issues of OFDM in underwater acoustic communications. Note that the performance of plain OFDM can be greatly improved via diversity combining through multiple receivers or linear precoding across the OFDM subcarriers due to the diversity benefits at the modulation level [9]. On the other hand, channel coding can improve OFDM performance through both diversity and coding benefits [9]. Since channel coding is indispensable in practical systems even with other diversity techniques, due to the coding gain, we adopt the coding-based approach in this paper. We propose to use nonbinary LDPC codes that match well the underlying modulation. We develop novel methods to construct nonbinary LDPC codes that have superior performance and are linear-time encodable in parallel. Based on the property that LDPC codes have high-density generator matrices, we show that the PAPR can be reduced considerably with minimal overhead through the selected mapping principle [10], [11]. This PAPR reduction approach requires multiple rounds of encoding for each information block at the transmitter, hence, the fast and parallel encoding algorithm for the proposed nonbinary LDPC codes is well suited.

We next survey current channel codes used in underwater acoustic communication, provide an overview on nonbinary LDPC codes, and then specify the contributions of this paper.

A. Current coding schemes used in underwater acoustic communications

Dedicated studies of coding for underwater acoustic communication are quite limited. A well-studied coding scheme from the existing literature is often used for underwater communication systems. For example, trellis coded modulation (TCM) was used with a classical adaptive receiver for single carrier transmissions in [12]. Convolutional codes and Reed Solomon (RS) codes have been tested in [13] for underwater acoustic communication. In conjunction with

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The authors are with the Department of Electrical and Computer Engineering, University of Connecticut, 371 Fairfield Way U-2157, Storrs, Connecticut 06269, USA (e-mail: {jhuang, shengli, willett}@engr.uconn.edu).

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spatial multiplexing, Turbo codes are used in [14] in a single-carrier underwater system with multiple transmitters. Space time trellis codes are used in [14] as well.

In an underwater acoustic OFDM system, serially concatenated convolutional codes have been tested with real data in [4]. However, only a non-iterative receiver was used, and the performance results are not impressive.

B. Nonbinary LDPC codes

Gallager's binary low-density parity-check (LDPC) codes [15] are excellent error-correcting codes that achieve performance close to the benchmark predicted by the Shannon theory [16]. Davey and Mackay first investigated the extension of LDPC to a nonbinary Galois field $\text{GF}(q)$ over the binary-input additive-white-Gaussian-noise (AWGN) channel [17]. It was shown empirically that nonbinary LDPC can potentially have better performance than binary irregular LDPC codes [17]. This has motivated active studies on nonbinary LDPC codes ever since.

The simplest LDPC codes are cycle codes [18], as their parity check matrices have column weight $j = 2$. An interesting finding in [19], [20] is that the mean column weight of nonbinary LDPC codes must approach 2 when the field order q increases; that is, the best nonbinary LDPC codes for very large q tend to be cycle codes over $\text{GF}(q)$. It has also been proved in [21] that cycle $\text{GF}(q)$ codes can achieve near-Shannon-limit performance as q increases. Further, numerical results in [21] demonstrate that cycle $\text{GF}(q)$ codes can outperform other LDPC codes, including degree-distribution-optimized binary irregular LDPC codes.

One main concern on nonbinary LDPC codes with large q is the decoding complexity. An FFT-based q -ary sum-product algorithm (FFT-QSPA) for decoding a general LDPC code over binary extension fields has been proposed in [22], [23], whose decoding complexity increases on the order of $O(q \log q)$. There also exists a min-sum version algorithm which works in the log-domain for nonbinary LDPC codes [24], similar to the min-sum decoding for binary LDPC codes where the Jaccobi operation $\max^*$ is replaced by the max operation. Reduced-complexity decoding algorithms for nonbinary LDPC codes have been recently developed in [25]–[28]. Using a geometrical vector representation and the table lookup, an efficient message-passing decoding algorithm for nonbinary LDPC codes over M -ary phase shift keying (PSK) has been developed in [25], which can perform close to the belief propagation decoding algorithm with far less decoding complexity. Truncating the size of extrinsic messages from q to n_m , the extended min-sum (EMS) algorithm in [26] reduces the total decoding complexity from the order of $O(q \log q)$ to $O(n_m \log n_m)$, where n_m could be much smaller than q . The improved version of the EMS algorithm in [27], [28] further reduces the message storage requirement.

One unique advantage of nonbinary LDPC codes over binary LDPC codes is that nonbinary codes can match very well the underlying modulation, and bypass the need for a symbol-to-bit conversion at the receiver. Reference [29] first applied nonbinary LDPC codes combined with high order modulation in a multi-input-multi-output (MIMO) communication system

with two transmitters and two receivers. A symbol-by-symbol joint demapper/decoding algorithm has been proposed which can greatly reduce the demapper complexity and results in a lower total complexity compared with the bit-by-bit joint demapper/decoding system proposed in [30]. It is shown in [31] through simulations that the iterative receivers with nonbinary LDPC codes over $\text{GF}(16)$ outperform the best optimized binary LDPC code in both performance and complexity, while a non-iterative system with regular LDPC cycle code over $\text{GF}(256)$ achieves the best performance with comparable decoding complexity as that of the binary iterative system [31]. Similar complexity and performance comparisons have been obtained in [32], [33] when LDPC cycle codes over $\text{GF}(16)$ and $\text{GF}(256)$ are combined with 16-QAM modulation in a MIMO system. The results in [29], [31]–[33] confirm the advantages of using nonbinary LDPC codes to match the underlying high order modulation.

C. Contribution and organization of this paper

This paper contains the following contributions.

- 1) We propose to apply nonbinary LDPC codes to match the underlying modulation in multicarrier underwater acoustic communications. We develop a code design procedure that leads to nonbinary LDPC codes with superior performance, while the encoding can be done in linear time and in parallel.
- 2) We use LDPC codes for PAPR reduction, based on their characteristic of having high-density generator matrices.
- 3) We present extensive simulation results that serve as benchmarks for future multicarrier modem design. We also include experimental results that demonstrate the effectiveness of nonbinary LDPC codes in real data.

The rest of the paper is organized as follows. We describe the system model in Section II, and present the proposed nonbinary LDPC codes in Section III. Using LDPC codes for PAPR reduction is presented in Section IV. Extensive simulation and experimental results are reported in Section V. Conclusions are drawn in Section VI.

II. SYSTEM MODEL

Fig. 1 shows the block diagram of an underwater OFDM system with nonbinary LDPC coding. Consistent with the block-by-block OFDM receiver in [7], encoding and decoding are done for each OFDM block separately.

Suppose that an LDPC code over $\text{GF}(q)$ is used where $q = 2^p$. Let $\{\alpha_0 = 0, \alpha_1, \dots, \alpha_{q-1}\}$ denote elements in $\text{GF}(q)$. Suppose that a constellation size of $M = 2^b$ will be used by the OFDM modulator. One advantage of nonbinary LDPC coding is that one can match the field order with the constellation size, i.e., $p = b$. This way, one element in $\text{GF}(q)$ is mapped to one point in the signal constellation. In some occasions when b is small, it may be preferable to choose $p > b$. Although other choices of p and b are possible, we present here the case with $J := p/b$ being an integer, for notational brevity. Each element in $\text{GF}(q)$ will then be mapped

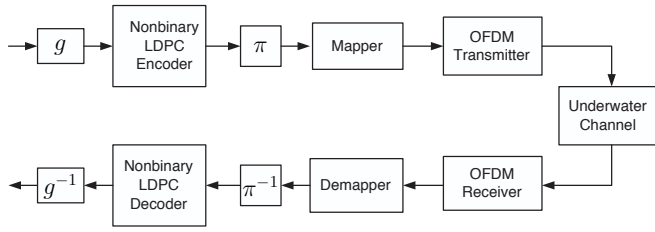


Fig. 1. A schematic block diagram of nonbinary LDPC coded OFDM system.

to J symbols drawn from the constellation. Let us describe the mapper as:

$$\alpha_i \longrightarrow [\phi^0(\alpha_i), \dots, \phi^{J-1}(\alpha_i)], \quad i = 0, \dots, q-1 \quad (1)$$

where $\phi^j(\alpha_i)$ is one point in the signal constellation.

Suppose that K_d OFDM subcarriers are used for data transmission, and the LDPC code rate is r . The transmitter operates as follows. For each OFDM block, rbK_d information bits are mapped to rbK_d/p symbols in $\text{GF}(q)$, with every p bits mapped to a single $\text{GF}(q)$ symbol through a bit-to-symbol mapper g . The LDPC encoder outputs bK_d/p coded symbols in $\text{GF}(q)$, which pass through a coded-symbol interleaver π to obtain a vector

$$\mathbf{u} = [u[0], \dots, u[K_d/J - 1]]^T. \quad (2)$$

The mapper in (1) maps the vector $\mathbf{u}$ to a modulated-symbol vector as

$$\begin{aligned} \mathbf{s} &:= [s[0], \dots, s[K_d - 1]]^T \\ &= [\phi^0(u[0]), \dots, \phi^{J-1}(u[0]), \\ &\quad \phi^0(u[1]), \dots, \phi^{J-1}(u[K_d/J - 1])]^T. \end{aligned} \quad (3)$$

The K_d entries of $\mathbf{s}$ are distributed to the OFDM data subcarriers. An OFDM block is formed after insertion of pilot and null subcarriers to data subcarriers; see more details in [7].

Using the block-by-block OFDM receiver, the equivalent channel input-output model on the data subcarriers is [7]:

$$y[k] = \hat{H}[k]s[k] + n[k], \quad k = 0, \dots, K_d - 1, \quad (4)$$

where $\hat{H}[k]$ is the estimated¹ channel frequency response on the k th data subcarrier, $y[k]$ is the output on the k th data subcarrier, and $n[k]$ is the composite noise with contributions from ambient noise, the residual inter-carrier interference (ICI), and the noise induced by channel estimation error. Assume that $n[k]$ has variance σ^2 per real and imaginary dimension. The average signal to noise ratio is defined as

$$E_s/N_0 = \frac{E_m \cdot E\{|\hat{H}[k]|^2\}}{2\sigma^2}, \quad (5)$$

where E_m is the average symbol energy of the constellation, $|\cdot|$ denotes the absolute value of a complex number, and $E\{\cdot\}$ denotes the expectation operation.

¹Note that the soft information or hard decisions from the channel decoder can be used to enhance the channel estimation accuracy. Such an iterative receiver structure is outside the scope of this paper.

When the noise variance σ^2 is available, the demapper can compute the likelihood

$$\begin{aligned} \Pr(u[k] = \alpha_i) \\ \propto \exp \left(\frac{-\sum_{j=0}^{J-1} |y[kJ + j] - \hat{H}[kJ + j]\phi^j(\alpha_i)|^2}{2\sigma^2} \right), \\ k = 0, \dots, K_d/J - 1; \quad i = 0, \dots, q-1. \end{aligned} \quad (6)$$

The likelihood values are passed to the deinterleaver π^{-1} before being passed to the LDPC decoder. The FFT-based q-ary sum-product algorithm [22], [23] can be used for iterative decoding. After a finite number of decoding iterations, hard decisions on the nonbinary symbols will be made at the output of the LDPC decoder, based on which information bits are found.

When the noise variance is not available, the demapper can compute the log-likelihood-ratio vector (LLRV) over $\text{GF}(q)$. The LLRV of $u[k]$ is defined as $\mathbf{z}[k] = [z_0[k], z_1[k], \dots, z_{q-1}[k]]^T$, where

$$z_i[k] = \ln \frac{\Pr(u[k] = \alpha_i)}{\Pr(u[k] = 0)}. \quad (7)$$

From equation (6), we have

$$\begin{aligned} z_i[k] = -\frac{1}{2\sigma^2} \sum_{j=0}^{J-1} \left(\left| y[kJ + j] - \hat{H}[kJ + j]\phi^j(\alpha_i) \right|^2 \right. \\ \left. - \left| y[kJ + j] - \hat{H}[kJ + j]\phi^j(0) \right|^2 \right). \end{aligned} \quad (8)$$

The LLRV values are passed to the deinterleaver π^{-1} before being passed to the LDPC decoder. The min-sum (MS) [24], or extended min-sum (EMS) algorithms [26]–[28] can be used for iterative decoding. Note that the LLRV generated by (8) is proportional to the reciprocal of σ^2 , and the updating rules of the MS (or EMS) decoding algorithm at the check nodes and variable nodes are linear operations with respect to the reciprocal of σ^2 [24]. Therefore, all the messages exchanged during decoding iterations are proportional to the reciprocal of σ^2 and the decoding results will remain unchanged with σ^2 set to an arbitrary value.

Note that when the code alphabet is matched to the modulation alphabet, i.e., $p = b$, or when p is an integer multiple of b , the interleaver in Fig. 1 is not necessary, as interleaving the coded symbols amounts to shuffling the columns of the parity check matrix of the LDPC code; hence interleaving can be absorbed into the code design. In such cases, the proposed system in Fig. 1 does not require any iterative processing between the demapper and the LDPC decoder regardless of the constellation labelling rules. This is because the demapper produces the likelihood probabilities (or LLRV) for each coded symbol over $\text{GF}(q)$ that are independent of other coded symbols. For other choices of p and b , interleaving and iterative demapping could be useful. Note that for a binary LDPC coded system with high order modulation, (i) other constellation labelling rules, e.g., set partitioning, can improve the system performance relative to Gray labelling, but require iterative processing between the maximum a posteriori (MAP)

demapper and the LDPC decoder [34], and (ii) the noise variance has to be estimated for demapping.

III. THE PROPOSED NONBINARY LDPC CODES

Let $\mathbf{H}$ denote the parity check matrix of an LDPC code, which has low density (or percentage) of nonzero entries. For binary codes over $\text{GF}(2)$, the nonzero entries of $\mathbf{H}$ can only have value 1. For nonbinary codes over $\text{GF}(q)$, the nonzero entries of $\mathbf{H}$ take values from the set $\{\alpha_1, \alpha_2, \dots, \alpha_{q-1}\}$. Suppose that the size of $\mathbf{H}$ is $m \times n$. An $n \times 1$ vector $\mathbf{x}$ with entries from $\text{GF}(q)$ is a valid codeword if and only if $\mathbf{H}\mathbf{x} = \mathbf{0}$. The column and row weights of $\mathbf{H}$ are defined as the number of nonzero entries in the column and row, respectively. An LDPC code whose $\mathbf{H}$ has fixed column weight and fixed row weight is called a regular code. An LDPC code whose $\mathbf{H}$ has fixed column weight $j = 2$ is called a cycle code [18], which could be regular or irregular.

In this paper, we will consider both regular and irregular LDPC codes over $\text{GF}(q)$. For regular LDPC codes, we will use cycle codes that have column weight 2 and fixed row weight d , based on the work in [35]. We will then develop a novel method to construct irregular LDPC codes that have mixed column weights of 2 and t , where $t \geq 3$.

A. Nonbinary regular cycle codes

LDPC cycle codes over $\text{GF}(2^p)$ can achieve near-Shannon-limit performance as p increases, as shown in [21]. Further, it is shown in [19], [20] that the column degree distribution of nonbinary LDPC code should be very sparse for large q . Therefore, the family of cycle codes are attractive when a large q is selected.

Using tools from graph theory, we have shown in [35] that the check matrix $\mathbf{H}$ of regular cycle codes is well structured. Specifically, the parity check matrix $\mathbf{H}$ of any regular cycle code can be put into a *concatenation form of row-permuted block-diagonal matrices* after row and column permutations if d is even, or, if d is odd and the code's associated graph contains at least one spanning subgraph that consists of disjoint edges [35]. For convenience, let us state Theorem 1 of [35] here, where the notation of $\mathbf{H}_1 \cong \mathbf{H}_2$ is used to denote the equivalence of two matrices $\mathbf{H}_1$ and $\mathbf{H}_2$ that one can be transformed to the other simply through row and column permutations.

Theorem 1 of [35]: For a regular cycle $\text{GF}(q)$ code with row weight $d = 2\nu$, its parity check matrix $\mathbf{H}$ of size $m \times n$ has the equivalent form

$$\mathbf{H} \cong [\tilde{\mathbf{H}}_1, \mathbf{P}_2 \tilde{\mathbf{H}}_2, \dots, \mathbf{P}_\nu \tilde{\mathbf{H}}_\nu], \quad (9)$$

where $\mathbf{P}_i$ is $m \times m$ permutation matrix, and $\tilde{\mathbf{H}}_i$ is of size $m \times m$, $1 \leq i \leq \nu$. The matrix $\tilde{\mathbf{H}}_i$ has an equivalent block-diagonal form

$$\tilde{\mathbf{H}}_i \cong \text{diag}(\tilde{\mathbf{H}}_{i,1}^c, \tilde{\mathbf{H}}_{i,2}^c, \dots, \tilde{\mathbf{H}}_{i,L_i}^c), \quad (10)$$

where the matrix $\tilde{\mathbf{H}}_{i,l}^c$ is of size $k_{i,l} \times k_{i,l}$ that satisfies $m = \sum_{l=1}^{L_i} k_{i,l}$ and has an equivalent form

$$\tilde{\mathbf{H}}^c = \begin{bmatrix} \zeta_1 & 0 & 0 & \dots & \beta_k \\ \beta_1 & \zeta_2 & 0 & \dots & 0 \\ 0 & \beta_2 & \zeta_3 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & \beta_{k-1} & \zeta_k \end{bmatrix} \quad (11)$$

with ζ_i s and β_i s being nonzero entries from $\text{GF}(q)$.

With the structure specified in (9) and (10), nonbinary regular cycle codes achieve several desirable properties [35]:

- 1) encoding can be done in linear-time and in parallel [35];
- 2) sequential belief propagation (BP) decoding can be implemented with parallel processing [35];

Three different methodologies have been presented to design good nonbinary regular cycle codes in [35], with one of them based on the equivalent form of $\mathbf{H}$.

Denote the designed code rate of an LDPC code with $\mathbf{H}$ of size $m \times n$ as $r = (n - m)/n$. Due to the constraint of $md = 2n$, the designed code rate of regular cycle codes is restricted to

$$r = \frac{n - m}{n} = \frac{d - 2}{d}, \quad (12)$$

where d is an integer. For example, r can be $\frac{1}{3}, \frac{1}{2}, \frac{3}{5}, \frac{2}{3}, \frac{5}{7}, \frac{3}{4}, \dots, \frac{7}{8}, \dots, \frac{15}{16}, \dots$

B. Nonbinary irregular LDPC code

Cycle codes over large Galois fields (e.g., $q \geq 64$) can achieve near-Shannon-limit performance [21]. However, cycle codes over small to moderate Galois fields (e.g., $4 \leq q \leq 32$) suffer from performance loss due to a “tail” in the low weight regime of the distance spectrum [21]. We thus resort to irregular weight distribution for nonbinary LDPC codes. Here we propose a simple strategy that improves the code performance while retaining the benefits of a regular cycle code as much as possible.

Our approach is to replace a portion of weight-2 columns of $\mathbf{H}$ of a cycle code by columns of weight $t > 2$, (e.g., $t = 3$ or $t = 4$). Let n_1 columns of $\mathbf{H}$ have weight 2 and n_2 columns have weight t . The mean column weight is

$$\eta = \frac{2n_1 + tn_2}{n} = 2 + (t - 2) \frac{n_2}{n} \quad (13)$$

In order to achieve linear-time encodability (this will be discussed in Section III-C), we restrict $n_1 \geq m$, that is, $0 \leq n_2 \leq (n - m)$. Therefore, we have $2 \leq \eta \leq 2 + (t - 2)r$ where $r = (n - m)/n$, and

$$n_1 = n \frac{(t - \eta)}{t - 2}, \quad n_2 = n \frac{(\eta - 2)}{t - 2}. \quad (14)$$

The matrix $\mathbf{H}$ can be arranged as

$$\mathbf{H} = [\mathbf{H}_1 \mid \mathbf{H}_2] \quad (15)$$

where $\mathbf{H}_1$ contains all weight-2 columns and $\mathbf{H}_2$ contains all weight- t columns. Clearly, $\mathbf{H}_1$ is of size $m \times n_1$ and $\mathbf{H}_2$ is of size $m \times n_2$.

Now we need to design $\mathbf{H}_1$ and $\mathbf{H}_2$. To maximally benefit from the structure of regular cycle codes presented in Section

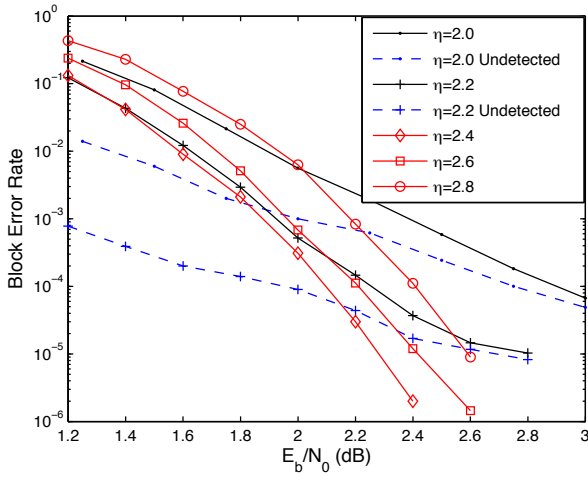


Fig. 2. Performance comparison of irregular codes over GF(16) with different mean column weight; $t = 3$, $r = 1/2$ and the codeword length is 1008 bits, i.e., 252 GF(16) symbols. The dashed lines depict the probability of undetected errors for the $\eta = 2.0$ and $\eta = 2.2$ cases.

III-A, we propose to use the following design rule. Note that $\mathbf{H}_1$ corresponds to the check matrix of a general cycle code. We would like $\mathbf{H}_1$ to be as close to a regular cycle code as possible. Specifically, we split the matrix as

$$\mathbf{H} = [\mathbf{H}_{1a} \mid \mathbf{H}_{1b} \mid \mathbf{H}_2] \quad (16)$$

where the matrix $\mathbf{H}_{1a}$ is of size $m \times n_{1a}$ and the matrix $\mathbf{H}_{1b}$ is of size $m \times n_{1b}$. The number n_{1a} is the largest integer not greater than n_1 that can render $d_{1a} = \frac{2n_{1a}}{m}$ an integer, that is, $\mathbf{H}_{1a}$ is the largest sub-matrix of $\mathbf{H}_1$ that could be made d_{1a} -regular. If $n_{1a} = n_1$, then $n_{1b} = 0$. As such, $\mathbf{H}_1$ itself can be made regular, which is a special case.

The detailed design steps are as follows.

- **Step 1: Specify the structure of $\mathbf{H}_{1a}$.**
Construct a cycle code of fixed row weight d_{1a} using the design methodologies presented in [35].
- **Step 2: Specify the structure of $\mathbf{H}_{1b}$ and $\mathbf{H}_2$.**
Apply the progressive-edge-growth (PEG) algorithm [36] to attach n_{1b} columns of weight 2 and n_2 columns of weight t to the matrix $\mathbf{H}_{1a}$. This way, the structure of $\mathbf{H}$ in (16) is established.
- **Step 3: Specify the non-zero entries of $\mathbf{H}_1$.**
Note that the sub-matrix $\mathbf{H}_1 = [\mathbf{H}_{1a} \mid \mathbf{H}_{1b}]$ can be regarded as a check matrix of a cycle code. Hence, we can apply the design criterion of [35] to choose appropriate nonzero entries for $\mathbf{H}_1$ to make as many as possible short length cycles of the associated graph [37] of $\mathbf{H}_1$ irresolvable.
- **Step 4: Specify the non-zero entries of $\mathbf{H}_2$.**
The nonzero entries of $\mathbf{H}_2$ are generated randomly with a uniform distribution over the set $\text{GF}(q) \setminus 0$.

The proposed nonbinary irregular LDPC codes try to make a large portion of the check matrix come from a regular cycle code. This way, many benefits of a regular cycle code can be retained.

As illustration, Fig. 2 compares the performance of irregular LDPC codes over GF(16) with different mean column weights.

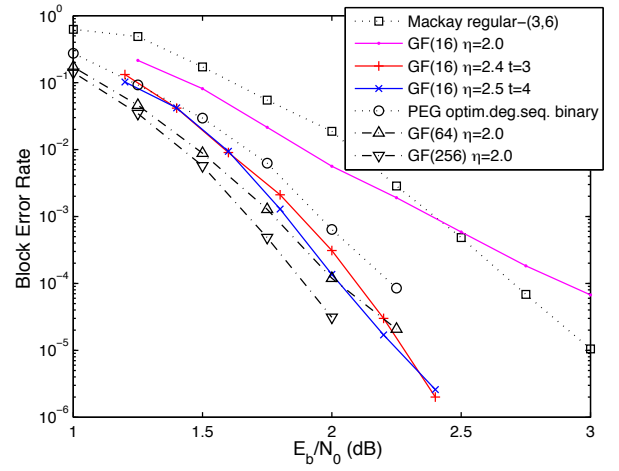


Fig. 3. Performance comparison of irregular codes over GF(16) with an optimized binary irregular LDPC code; $r = 1/2$ and the codeword length is 1008 bits, which correspond to 252 GF(16) symbols, 168 GF(64) symbols, and 126 GF(256) symbols.

All the codes have rate $1/2$ and codeword length of 1008 bits that correspond to 252 GF(16) symbols. BPSK modulation is used on the binary input AWGN channel and the decoder uses the sequential BP algorithm [38], [39] with a maximum of 80 iterations. We observe from Fig. 2 that the performance curves of the codes with $\eta = 2.0$ and $\eta = 2.2$ level off above 10^{-5} due to the contribution from the probability of undetected errors. This is not the case if $\eta \geq 2.4$. Actually no undetected errors have been observed for $\eta \geq 2.4$ in our simulations. Another interesting observation is that as η increases from 2.4 to 2.6 and 2.8, the code performance degrades. Therefore, the code with $\eta = 2.4$ is the best one for this particular example. Fig. 3 shows the performance comparison between the irregular LDPC codes over GF(16) and an optimized binary irregular LDPC code. The performance of Mackay's (3,6)-regular code and cycle codes over GF(64) and GF(256) of the same block length are also included (see more details on these codes' parameters in [35]). It can be seen from Fig. 3 that by adopting an irregular column weight distribution, the code's performance can be greatly improved without having to use a very large q ; this is desirable from both the complexity and the constellation matching perspectives.

C. Linear-time encoding in parallel

Decoding of LDPC codes can use standard sum-product algorithms and its low-complexity variants, while encoding has been an important issue for LDPC codes. Encoding of a general LDPC code can be done in almost linear time instead of quadratic time on the block length [40].

The proposed codes in Section III-B can be encoded in linear time in parallel as follows. Assume $n_1 \geq m$, then $n_{1a} \geq m$. Since $\mathbf{H}_{1a}$ is regular, it can be decomposed as in (9). Hence, the first $m \times m$ submatrix of $\mathbf{H}$ can be made to have the form in (10). Let us denote it as $\bar{\mathbf{H}}_1$, and split $\mathbf{H}$ as $\mathbf{H} = [\bar{\mathbf{H}}_1 \mid \mathbf{H}']$ where $\mathbf{H}'$ is of size $m \times (n - m)$. We will make sure that $\bar{\mathbf{H}}_1$ has full rank. Partition the codeword $\mathbf{x}$ into two parts as $\mathbf{x} = [\mathbf{p}^T, \mathbf{s}^T]^T$ where $\mathbf{p}$ is of length m . Let $\mathbf{p}$ contain

the parity symbols and $\mathbf{s}$ contain the information symbols. A valid codeword satisfies $\mathbf{H}\mathbf{x} = \mathbf{0}$, which implies that

$$\bar{\mathbf{H}}_1 \mathbf{p} = -\mathbf{H}' \mathbf{s}. \quad (17)$$

From (10), $\bar{\mathbf{H}}_1$ is block diagonal $\text{diag}(\tilde{\mathbf{H}}_{1,1}^c, \dots, \tilde{\mathbf{H}}_{1,L_1}^c)$. According to the sizes of $\{\tilde{\mathbf{H}}_{1,l}^c\}_{l=1}^{L_1}$, let us partition $\mathbf{p}$ and the right-hand side of (17) into L_1 pieces as

$$\mathbf{p} = [\mathbf{p}_1^T, \dots, \mathbf{p}_{L_1}^T]^T, \quad (18)$$

$$-\mathbf{H}' \mathbf{s} = [\mathbf{b}_1^T, \dots, \mathbf{b}_{L_1}^T]^T, \quad (19)$$

respectively. Computation of $\mathbf{p}$ requires solving the following L_1 equations

$$\tilde{\mathbf{H}}_{1,i}^c \mathbf{p}_i = \mathbf{b}_i, \quad 1 \leq i \leq L_1. \quad (20)$$

A linear time algorithm for solving these equations has been proposed in Lemma 4 of [37]. Specifically, to solve an equation in the form of $\tilde{\mathbf{H}}^c \mathbf{x} = \mathbf{b}$, where $\mathbf{x} = [x_1, x_2, \dots, x_k]^T$, $\mathbf{b} = [b_1, b_2, \dots, b_k]^T$, and $\tilde{\mathbf{H}}^c$ has the structure in (11), we can use the following algorithm [37].

1. $z_1 = b_1; z_i = \gamma_{i-1} z_{i-1} + b_i, i = 2, 3, \dots, k;$
2. $y_k = (1 + \gamma_1 \gamma_2 \dots \gamma_k)^{-1} z_k;$
 $y_i = z_i - \gamma_1 \gamma_2 \dots \gamma_{i-1} \gamma_k y_k, i = 1, 2, \dots, k-1;$
3. $x_i = \zeta_i^{-1} y_i, i = 1, 2, \dots, k.$
 where $\gamma_i = \zeta_i^{-1} \beta_i, i=1, 2, \dots, k.$

Assume that the coefficients have been stored before computing. The computation complexity is $2(k-1)$ additions, $2(k-1)$ multiplications, and $k+1$ divisions over $\text{GF}(q)$.

Note that solving those L_1 equations in (20) can be done in parallel. The overall complexity of solving the equation (17) is about $2m$ additions, $2m$ multiplications, and m divisions over $\text{GF}(q)$, when some coefficients are precomputed. Note that the universal linear-time encoding algorithm of [37] for cycle codes works only in a serial manner. Fast and parallel encoding is quite desirable especially when the block length is large, or, when multiple rounds of encoding is needed for the proposed OFDM PAPR reduction as will be detailed in the next section.

IV. PEAK-TO-AVERAGE POWER RATIO REDUCTION

One major issue of OFDM is the high peak-to-average power ratio (PAPR), which is defined as

$$\text{PAPR} := \frac{\max(|x(t)|^2)}{E[|x(t)|^2]}, \quad (21)$$

where $x(t)$ is the transmitted OFDM signal. PAPR can be evaluated at either baseband or passband, depending on the choice of $x(t)$ [41]. Nonlinear amplification causes inter modulation among subcarriers and undesired out-of-band radiation. To limit nonlinear distortion, the amplifier at the transmitter must operate with large power back-offs.

Various PAPR reduction methods have been proposed for radio OFDM systems [41]. In this paper, we adopt the selected mapping (SLM) approach in [10], [11]. In SLM, the transmitter generates a set of sufficiently different candidate signals which all represent the same information and selects the one with the lowest PAPR for transmission. In the original SLM approach [10], side information on which signal candidate has

been chosen needs to be transmitted. This causes signalling overhead. In addition, side information has high importance and has to be strongly protected. In the modified approach [11], some additional bits, used to select different scrambling code patterns, are inserted to the information bits, before scrambling and channel encoding take place. This way, the side information bits are contained in the data and do not need separate encoding.

The fact that the generator matrix $\mathbf{G}$ of an LDPC code has high density is well known [40], but rarely utilized. Here we use this property of LDPC to reduce PAPR, following the SLM principle in [11]. Such an application of LDPC code structure is novel and has not been reported elsewhere to our knowledge. The transmitter operates as follows:

- For each set of information bits to be transmitted within one OFDM block, reserve z bits for PAPR reduction purposes.
- For each choice of the values of these z bits, carry out LDPC encoding and OFDM modulation, and calculate the PAPR.
- Out of 2^z candidates, select the OFDM symbol with the lowest PAPR for transmission.

Compared with [11], the proposed method bypasses the scrambling operation at the transmitter and the descrambling operation at the receiver. Due to the non-sparseness of $\mathbf{G}$, a single bit change will lead to a drastically different codeword after LDPC encoding [40]. Since z is very small, e.g., $z = 2$, or $z = 4$, the reduction on transmission rate is negligible. At the receiver side, those z bits are simply dropped after channel decoding. The main complexity increase is at the transmitter. Fast encoding as presented in Section III-C is thus very desirable.

For a systematic nonbinary LDPC code with size $n \times k$, there is a $k \times k$ identity matrix contained in $\mathbf{G}$. Every information bit change can only cause significant changes on the $(n-k)$ parity symbols. For low rate transmissions, systematic LDPC may achieve decent PAPR reduction. However, for high rate transmissions where $(n-k)$ is small, nonsystematic LDPC codes may be preferred over systematic codes for PAPR reduction.

One simple way to construct a nonsystematic code from a systematic code is as follows. Put the z reserved bits into the last s information symbols of the block $\mathbf{u}$, where $s = \lceil z/p \rceil$. Construct a matrix $\mathbf{V}$ as

$$\mathbf{V} = \begin{bmatrix} \mathbf{I}_{k-s} & \mathbf{B} \\ \mathbf{0} & \mathbf{A} \end{bmatrix} \quad (22)$$

where $\mathbf{A}$ is an invertible square matrix of size $s \times s$ and $\mathbf{B}$ is of size $(k-s) \times s$. Then construct the generator matrix of the nonsystematic code from that of a systematic code as

$$\mathbf{G}_{\text{non}} = \mathbf{G}_{\text{sys}} \mathbf{V}. \quad (23)$$

The output codeword is then $\mathbf{x} = \mathbf{G}_{\text{non}} \mathbf{u} = \mathbf{G}_{\text{sys}} \mathbf{V} \mathbf{u}$, which means that the information block $\mathbf{u}$ is scrambled by the matrix $\mathbf{V}$ before being passed to the systematic encoder. At the decoder, we first recover an estimate of $\mathbf{V} \mathbf{u}$, and then obtain $\hat{\mathbf{u}}$ by left multiplying the inverse of $\mathbf{V}$ as

$$\mathbf{V}^{-1} = \begin{bmatrix} \mathbf{I}_{k-s} & -\mathbf{B}\mathbf{A}^{-1} \\ \mathbf{0} & \mathbf{A}^{-1} \end{bmatrix}. \quad (24)$$

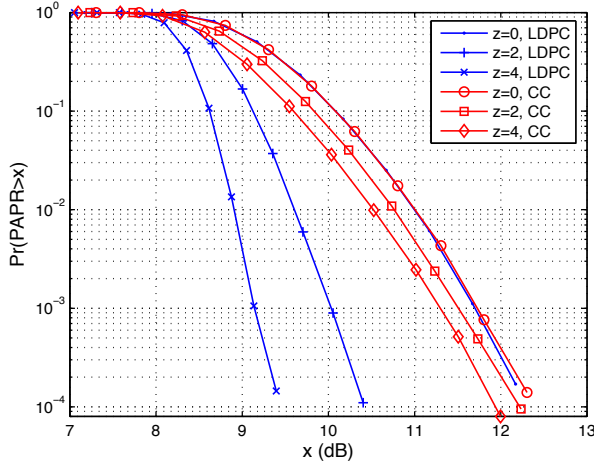


Fig. 4. Comparison of PAPR reduction using LDPC and CC codes. Code rate is 1/2.

Note that the size of $\mathbf{A}$ is very small. For example, if $z = 4$, then $s = 2$ when using an LDPC code over GF(4), and $s = 1$ when using an LDPC code over GF(16). Therefore, left multiplication of $\mathbf{V}^{-1}$ has low complexity and can be done in parallel.

With the OFDM parameters as in [7], [8], where each OFDM block has 1024 subcarriers out of which 672 subcarriers are used for data transmission, we simulate the baseband OFDM signals with a sampling rate 4 times of the bandwidth to evaluate the complementary cumulative distribution function (ccdf), $\Pr(\text{PAPR} > x)$. The PAPR ccdf curves for mode 2 of Table I are shown in Fig. 4 for $z = 0$, $z = 2$, and $z = 4$, respectively, where the corresponding curves using a 64-state rate-1/2 convolutional code (with generators (133, 171)) are also included. The generator matrix of convolutional code has low density, as each bit can only affect subsequent bits within the constraint length. For convolutional codes, the z reserved bits are distributed uniformly among the information bit sequence. We observe from Fig. 4 that using a nonbinary LDPC code with 4 bits overhead can achieve about 3dB gain than the case with no overhead at the ccdf value of 10^{-3} . Compared with convolutional codes using 4 bits overhead, nonbinary LDPC code with 4 bits overhead can achieve about 2dB gain at the ccdf value of 10^{-3} . Scrambling can be used together with convolutional codes to improve the PAPR characteristic [11]. But this is not necessary with LDPC codes. With rate 1/2, we find that systematic and nonsystematic codes have similar PAPR reduction performance. Fig. 5 shows that nonsystematic LDPC codes have better PAPR reduction than systematic codes when the code rate is increased to 3/4.

V. PERFORMANCE RESULTS

In this section, we present both Monte Carlo simulation results and field test results.

A. Monte Carlo simulation results

We simulate the system performance using two channel models.

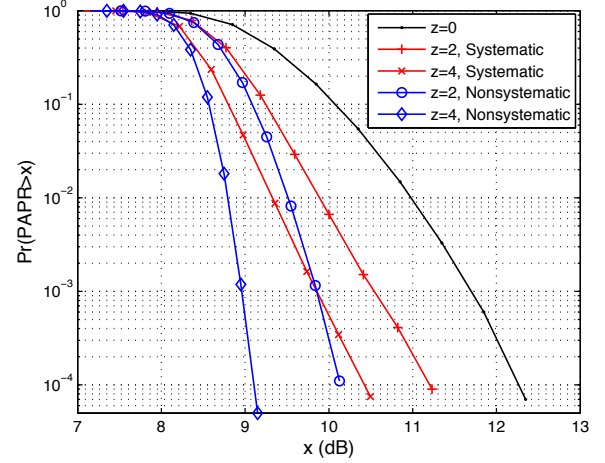


Fig. 5. Comparison of PAPR reduction using systematic and nonsystematic LDPC codes. Code rate is 3/4.

- 1) an additive white Gaussian noise channel, that is, $\hat{H}[k] = 1, \forall k$, in (4).
- 2) a Rayleigh fading multipath channel with bandwidth 12 kHz and delay spread of 10 ms, resulting in 120 channel taps in discrete-time. We assume all the channel taps are complex Gaussian random variables with equal variance.

The two channel models are drastically different, one without channel fading and the other with multipath fading from a rich-scattering environment. We compare the coding performance based on these two different channel models to facilitate code selection. Note that practical underwater acoustic channels could be far more complex, e.g., with sparse multipath structure and much longer impulse response.

When the LDPC coding alphabet is matched to the modulation alphabet, i.e., $p = b$, or when p is an integer multiple of b , constellation labelling does not affect the error performance of the proposed system. Further, interleaving the codeword means a column rearrangement of the code's parity check matrix, implying that interleaving can be absorbed into the code design and does not need to be considered explicitly. In the following simulation results, Gray labelling and identity interleavers are used.

We use the OFDM parameters as in [7] and [8]. Each OFDM block is of duration 85.33 ms, and has 1024 subcarriers, out of which 672 subcarriers are used for data transmission. Each OFDM block contains one codeword. The FFT-QSPA algorithm [22] is used for nonbinary LDPC decoding, where the maximum number of iterations is set to 80.

Test Case 1 (Combination of coding and modulation). Figs. 6 and 7 compare the error performance of different coding and modulation combinations under the AWGN and Rayleigh fading channels, respectively. We have the following observations.

- A QPSK system with rate 7/8 coding over GF(16) leads to a data rate of 1.75 bits/symbol while a 16-QAM system with rate 1/2 coding over GF(16) and an 8-QAM system with rate 2/3 coding over GF(8) lead to a data rate of 2 bits/symbol. (The 8-QAM constellation used in this paper

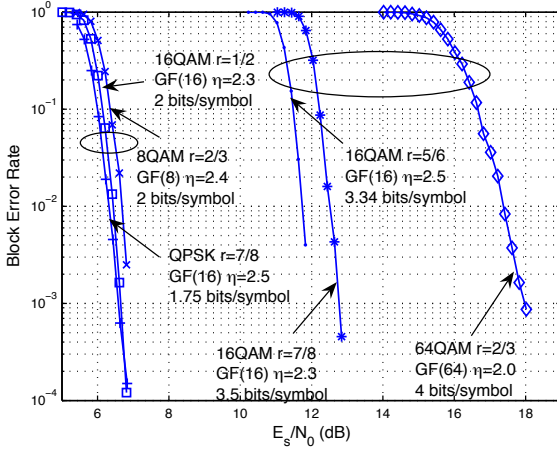


Fig. 6. Performance comparison of different coded modulation schemes over the AWGN channel.

is from [42, Fig. 4.3-4]). Over the AWGN channel, these three systems achieve similar performance as shown in Fig. 6. However, the QPSK system with rate 7/8 coding (the 8-QAM system with rate 2/3 coding, resp.) is about 4dB (1.3dB, resp.) worse than the 16-QAM system with rate 1/2 over the Rayleigh fading channel at BLER of 10^{-2} , as seen from Fig. 7.

- A 64-QAM system with rate 2/3 coding has data rate of 4 bits/symbol, while a 16-QAM with rate 5/6 (7/8, resp.) coding has data rate of 3.34 (3.5, resp.) bits/symbol. Over the AWGN channel, the 16-QAM system with rate 5/6 coding (16-QAM system with rate 7/8 coding, resp.) achieves about 5.7dB (5dB, resp.) gain against the 64-QAM system with rate 2/3 coding at BLER of 10^{-2} , as seen from Fig. 6. However, the 16-QAM system with rate 5/6 coding has similar performance as the 64-QAM system with rate 2/3 coding over the Rayleigh fading channel, and the 16-QAM system with rate 7/8 coding is about 2dB worse than the 64-QAM system with rate 2/3 coding over the Rayleigh fading channel at BLER of 10^{-2} , as seen from Fig. 7.

Hence, different coding and modulation combinations with a similar data rate could have quite different behaviors in the AWGN and Rayleigh fading channels. This is due to the fact that different performance metrics matter for AWGN and Rayleigh fading channels, as observed in [43]. Specifically, minimum Hamming distance plays a significant role for the Rayleigh fading channel while minimum Euclidean distance plays a significant role for the AWGN channel [43]. In general, a combination of low rate code and large constellation yields a larger Hamming distance than that of high rate code and small constellation, when the same spectral efficiency is achieved.

We have simulated the performance of many different combinations of modulations such as BPSK, QPSK, 8-QAM, 16-QAM and 64-QAM, and LDPC codes of rate 1/2, 2/3, 3/4, 5/6 and 7/8. For LDPC codes over $GF(q)$ where $q < 64$, different combinations of value t (3 or 4) and η (range from 2.0 to 3.0) have been simulated. For LDPC codes over $GF(64)$, regular cycle codes from [35] are used. For the bandwidth

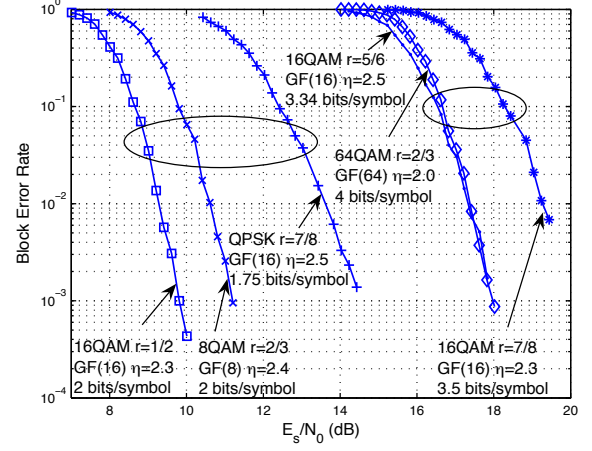


Fig. 7. Performance comparison of different coded modulation schemes over the Rayleigh fading channel.

TABLE I
NONBINARY LDPC CODES DESIGNED FOR UNDERWATER SYSTEM. η STANDS FOR MEAN COLUMN WEIGHT. EACH CODEWORD HAS $672b$ BITS WITH A SIZE- 2^b CONSTELLATION.

Mode	Bits Per Symbol	Code Rate	η	t	Galois Field	Constellation
1	0.5	1/2	2.8	4	GF(4)	BPSK
2	1	1/2	2.8	4	GF(4)	QPSK
3	1.5	1/2	2.8	4	GF(8)	8-QAM
4	2	1/2	2.3	3	GF(16)	16-QAM
5	3	1/2	2.0	-	GF(64)	64-QAM
6	4	2/3	2.0	-	GF(64)	64-QAM
7	5	5/6	2.0	-	GF(64)	64-QAM

efficiency ranging from 0.5 to 5 bits/symbol, we only keep the combination that has good performance in the Rayleigh fading channel and record the LDPC code parameters. We collect the seven modulation-coding pairs in Table I, and propose to use these seven modes for future OFDM modem development. It can be seen from Table I that low-rate codes (i.e., rate 1/2) are preferable.

Test Case 2 (Performance of different modes). Figs. 8 and 9 show the block-error-rate (BLER) and bit-error-rate (BER) performance of all the modes in Table I over the AWGN channel. Figs. 10 and 11 show the BLER and BER performance of all the modes in Table I over the Rayleigh fading channel.

In Figs. 8–11, we also plot the uncoded BER curves corresponding to the constellations used. We see that as long as the uncoded BER is somewhat below 0.1, the coding performance improves drastically. Indeed, the performance curves are very steep in the *waterfall* region.

Test Case 3 (Comparison with convolutional codes based BICM). Figs. 12 and 13 show the performance comparisons between a bit-interleaved coded-modulation (BICM) system [44], [45] based on a 64-state rate-1/2 convolutional code with the generator (133,171) and the proposed nonbinary LDPC coding system over the AWGN and Rayleigh fading channels,

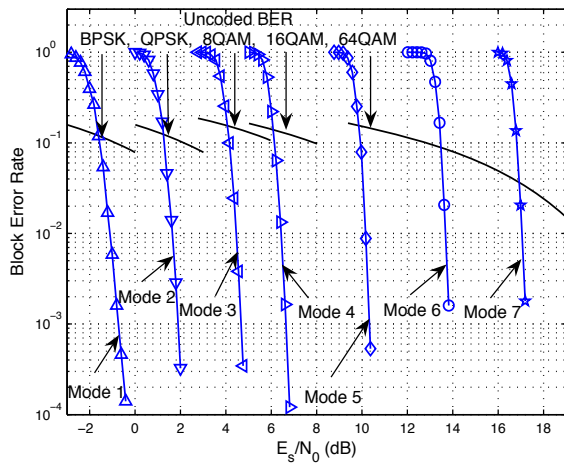


Fig. 8. BLER performance of different modes over the AWGN channel.

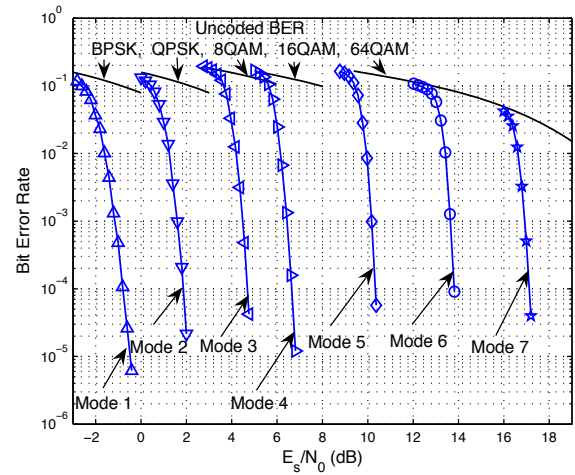


Fig. 9. BER performance of different modes over the AWGN channel.

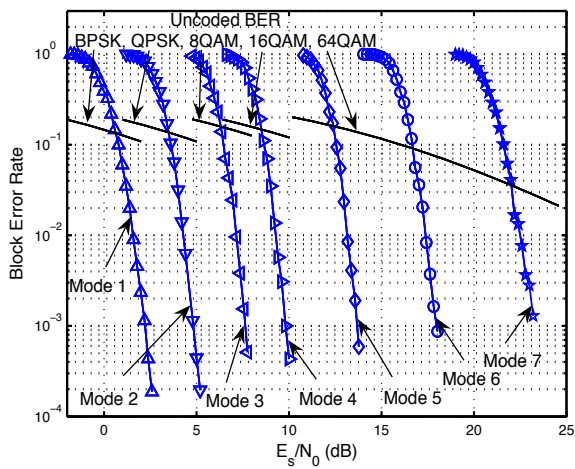


Fig. 10. BLER performance of different modes over the Rayleigh fading channel.

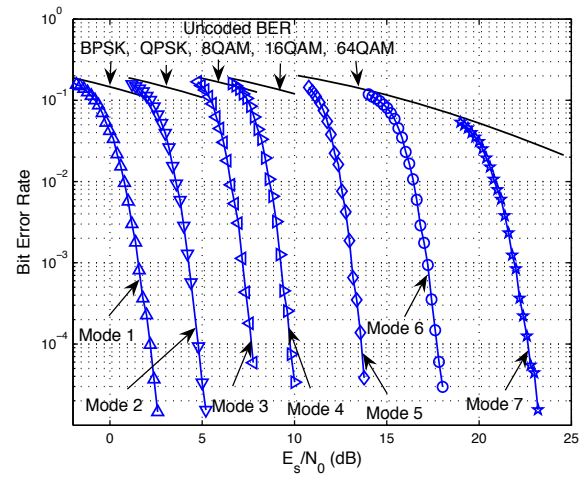


Fig. 11. BER performance of different modes over the Rayleigh fading channel.

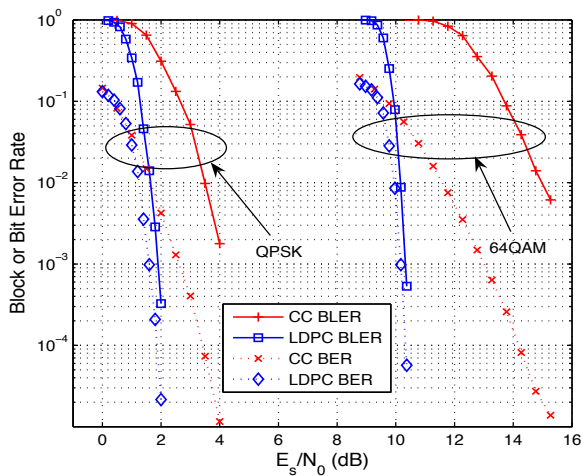


Fig. 12. Comparison between LDPC and convolutional codes of rate 1/2 under different modulation over the AWGN channel.

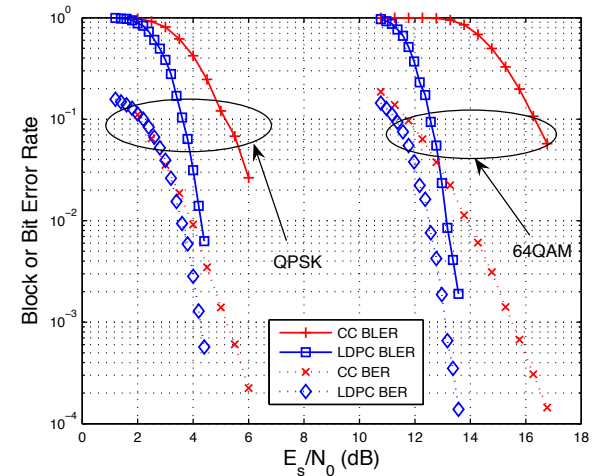


Fig. 13. Comparison between LDPC and convolutional codes of rate 1/2 under different modulation over the Rayleigh fading channel.

TABLE II
BER RESULTS FOR CC WITH QPSK

Bandwidth B	1 receiver uncoded / coded	2 receivers uncoded / coded
AUV Fest, 3 kHz	0.1219 / 0.0403	0.0395 / 0
AUV Fest, 6 kHz	0.0762 / 0.0063	0.0218 / 0
AUV Fest, 12 kHz	0.0752 / 0.0048	0.0185 / 0
Bay Test, 25 kHz	0.0016 / 0	-
Bay Test, 50 kHz	0.0834 / 0.0191	0.0243 / 0

TABLE III
BER RESULTS FOR LDPC WITH QPSK

Bandwidth B	1 receiver uncoded / coded	2 receivers uncoded / coded
AUV Fest, 12 kHz	0.0613 / 0	-
Bay test, 25 kHz	0.0015 / 0	-
Bay test, 50 kHz	0.1828 / 0.1851	0.1102 / 0

respectively. Gray labelling, random bit-level interleaver, and soft decision Viterbi decoding are used in the BICM system. We observe from Figs. 12 and 13 that compared with the BICM system using the convolutional code, nonbinary LDPC codes achieve several decibels (varying from 2 to 5 dB) performance gain at BLER of 10^{-2} .

We underscore that the performance of BICM could be considerably improved by using more powerful binary codes such as turbo codes and binary LDPC codes, and through iterative constellation demapping [46]. Further comparisons of the proposed nonbinary LDPC coded system with various BICM variants are needed for future work.

B. Field test results from experiments at AUV Fest 2007 and Buzzards Bay, 2007

We have applied nonbinary LDPC codes in a multicarrier system and collected data from experiments at AUV Fest, Panama City, FL, June 2007, and at Buzzards Bay, MA, Aug. 2007; see [2] for detailed descriptions.

In the AUV Fest, the sampling rate was 96 kHz. We used signals with three different bandwidths, 3 kHz, 6 kHz, and 12 kHz, centered around the carrier frequency 32 kHz. The transmitter was about 9 m below a surface buoy. The receiving boat had an array in 20m-depth water. The array depth was 9 m to the top of cage. In this paper, we report the results with a transmission distance of 500 m. The channel delay spread is about 18 ms [2].

In the Buzzards Bay test, the sampling rate was 400 kHz. We used signals with two different bandwidths, 25 kHz and 50 kHz, centered around the carrier frequency 110 kHz. The transmitter gear was deployed to the depth of 6 m to 7.6 m with water depth 14.3 m. Receiver array was deployed to the depth of 6m with water depth 14.3 m. Array spacing is 0.2 m. We report here the results with a transmission distance of 180 m. The channel delay spread is about 2.5 ms [2].

In both experiments, mode 2 (QPSK) and mode 4 (16-QAM) listed in Table I were adopted for nonbinary LDPC coding. In addition, we included signal sets with convolutional coding, where a 16-state rate 1/2 convolutional code with the generator polynomial (23,35) is used.

With QPSK modulation and rate 1/2 coding, the achieved spectral efficiency after accounting for various overheads is

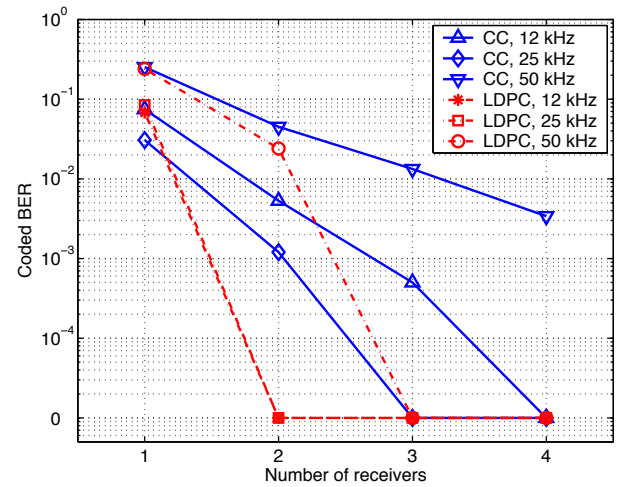


Fig. 14. The coded BER with 16-QAM constellation and rate 1/2 coding. The results with the 12 kHz bandwidth are from the AUV Fest experiment, while the results with the bandwidths of 25 kHz and 50 kHz are from the Buzzards Bay experiment.

about 0.5 bits/sec/Hz, leading to data rates from 1.5 kbps to 25 kbps with different bandwidths from 3 kHz to 50 kHz [2]. With 16-QAM modulation and rate 1/2 coding, the achieved spectral efficiency is about 1 bits/sec/Hz, leading to data rates from 12 kbps to 50 kbps with different bandwidths from 12 kHz to 50 kHz [2].

1) *BER performance for QPSK*: BER results for convolutional codes (CC) with QPSK are collected in Table II, and those for LDPC codes are in Table III. A total of 43008 information bits were transmitted in each setting.

In some cases, there is no decoding error even with a single receiver. When signals from two receivers are properly combined, there is no error after channel decoding for all the cases tested.

2) *BER performance for 16-QAM*: The BERs after channel decoding are plotted in Fig. 14, when 16-QAM is used. A total of 43008 information bits were transmitted in each setting. For the $B = 12$ kHz case from the AUV Fest experiment, two receivers are needed for zero BER for LDPC, while four receivers are needed for zero BER for CC. For the $B = 25$ kHz case from the Buzzards Bay test, two receivers are needed for zero BER for LDPC, while three receivers are needed for zero BER for CC. For the $B = 50$ kHz case from the Buzzards Bay test, three receivers are needed for zero BER for LDPC, while for CC, a large BER still occurs with four receivers. This is because the LDPC code has much better error-correction capability than the convolutional code used.

C. Field test results from the RACE08 experiment

The Rescheduled Acoustic Communications Experiment (RACE) took place in Narragansett Bay, Rhode Island, from March 1st through March 17th, 2008. The water depths in the area ranges from 9 to about 14 meters. The primary source for acoustic transmissions was located approximately 4 meters above the bottom. Three receiving arrays, one at 400 meters to the east from the source, one at 400 meters to the north

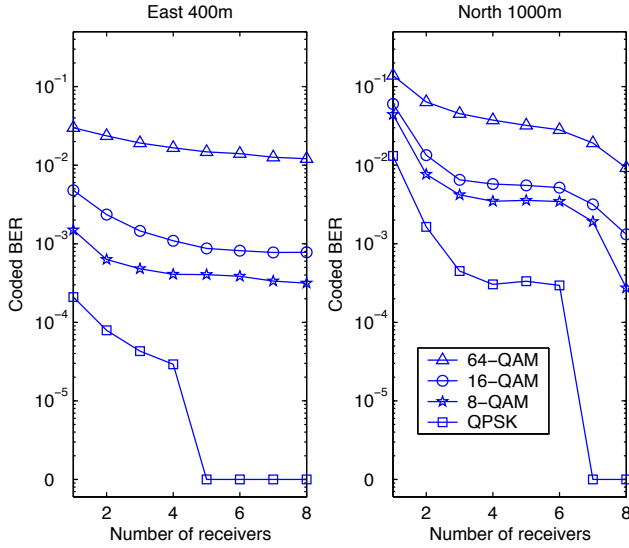


Fig. 15. The coded BER as a function of the number of receive-elements; the results are averaged over the data collected from 13 days in the RACE08 experiment.

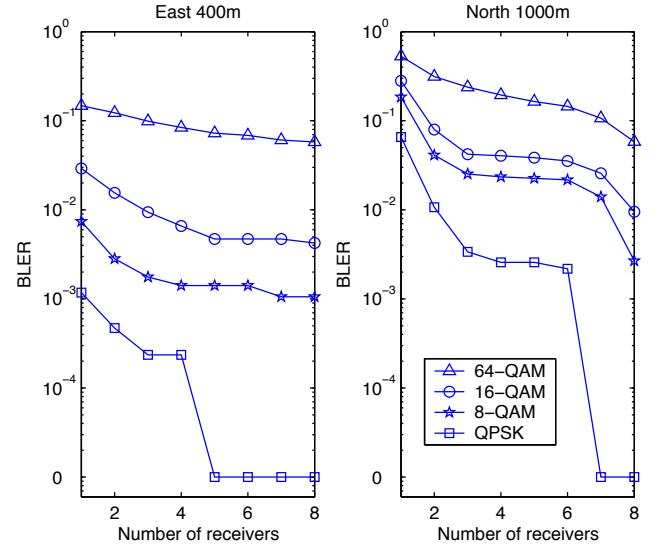


Fig. 16. The block error rate as a function of the number of receive-elements; the results are averaged over the data collected from 13 days in the RACE08 experiment.

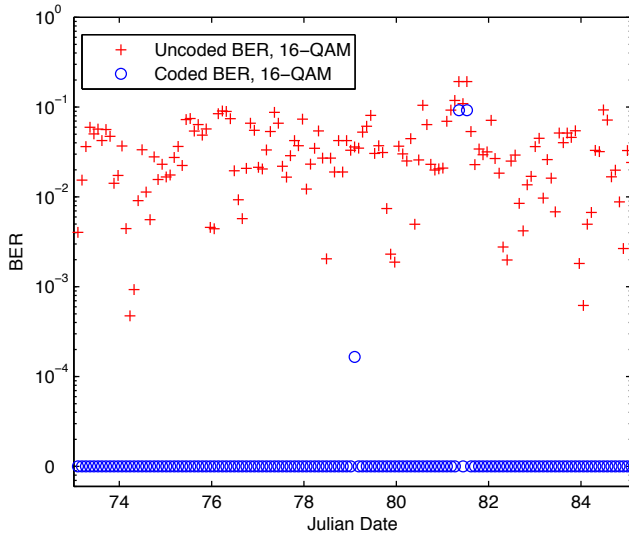


Fig. 17. Bit error rates in different Julian dates, North 1000m, 8 receive-elements, 16-QAM

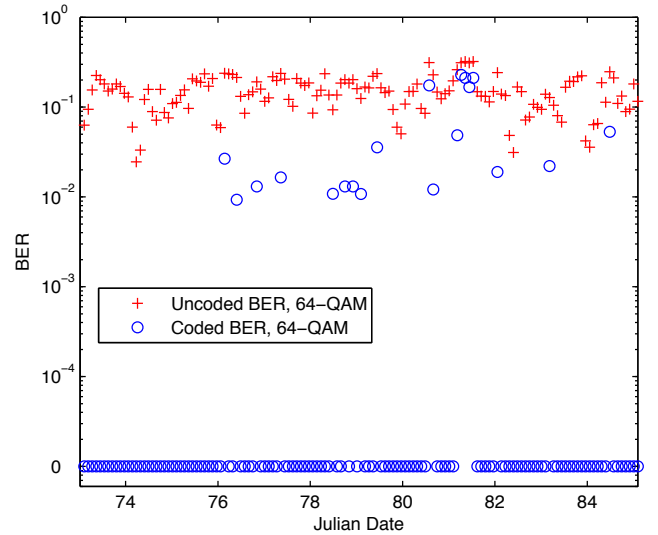


Fig. 18. Bit error rates in different Julian dates, North 1000m, 8 receive-elements, 64-QAM

from the source, and one at 1000 meters to the north from the source, were located with the bottom of the arrays about 2 meters above the sea floor. The arrays at 400 meters range are 24 element vertical arrays with a spacing of 5 cm between elements. The array at the 1000 meter range is a 12 element vertical array with 12 cm spacing between elements.

The sampling rate was $f_s = 39.0625$ kHz. We set the signal bandwidth as $B = f_s/8 = 4.8828$ kHz, centered around the carrier frequency $f_c = 11.5$ kHz. We use $K = 1024$ subcarriers, which leads to a subcarrier spacing of $\Delta f = 4.8$ Hz and the OFDM duration of $T = 209.7152$ ms. The guard interval between consecutive OFDM blocks is $T_g = 25$ ms.

We test the transmission modes 2 to 5 listed in Table I. Our transmission file contains four packets. The first packet contains 36 OFDM blocks with QPSK modulation (Mode 2),

the second packet contains 24 OFDM blocks with 8-QAM modulation (Mode 3), the third packet contains 18 OFDM blocks with 16-QAM modulation (Mode 4), and the last packet contains 12 OFDM blocks with 64-QAM modulation (Mode 5). Each packet has 24192 information bits regardless of the transmission modes. Accounting for the overheads of guard interval insertion, channel coding, pilot and null subcarriers, the spectral efficiency is:

$$\beta = \frac{T}{T + T_g} \cdot \frac{672}{1024} \cdot \frac{1}{2} \cdot \log_2 M \text{ bits/sec/Hz.} \quad (25)$$

The spectral efficiencies for the RACE08 experiment are then 0.5864, 0.8795, 1.1727, and 1.7591 bits/sec/Hz, for transmission modes with QPSK, 8-QAM, 16-QAM, and 64-QAM constellations, respectively. The achieved data rates are 2.86, 4.29, 5.72, and 8.59 kbps, respectively.

During the experiment, each transmission file was transmitted twice every four hours, leading to 12 transmissions each day. A total of 124 data sets were successfully recorded on each array within 13 days from the Julian date 073 to the Julian date 085.

Due to space limitations, we here report the performance results on the array at 400m to the east, and on the array at 1000m to the north. The channel delay spreads are around 5 ms for both settings. Figs. 15 and 16 depict the BER and BLER after channel decoding as a function of the number of receive-elements, averaged over all the data sets collected from 13 days. Hence, each point in Figs. 15 and 16 corresponds to transmissions of $124 \times 24192 \approx 3.0 \cdot 10^6$ information bits. Figs. 17 and 18 plot the uncoded and coded BERs for each recorded data set at the array at 1000m to the north across the Julian dates, for 16-QAM and 64-QAM constellations, respectively. We observe that with 8 receive-elements, *error-free* performance was achieved during the 13 day operation for QPSK transmissions. Very good performance was achieved for 8-QAM and 16-QAM transmissions, as the BLER is below 10^{-2} , which may satisfy the requirement of a practical system. The average BLER is below 0.1 for 64 QAM constellation. A closer look at Fig. 18 shows that error-free transmissions were achieved for a large majority of transmissions. Results from this experiment demonstrate that the proposed transmission modes are fairly robust to the varying channel conditions within those 13 days.

In summary, we have applied nonbinary LDPC coding in multicarrier underwater systems, where we focus on the case of matching the code alphabet with the modulation alphabet. Our experience with real data is that whenever the uncoded BER is below 0.1, normally no decoding errors will occur for the rate 1/2 nonbinary LDPC codes used. This is consistent with the simulation results in Figs. 8–11, as the curves at the waterfall region are steep. The uncoded BER can serve as a quick performance indicator to assess how likely the decoding will succeed. We argue that *the goal of an OFDM receiver design is to achieve an uncoded BER within the range of 0.1 and 0.01, as nonbinary LDPC coding will boost the overall system performance afterwards.*

VI. CONCLUSION

In this paper, we proposed the use of nonbinary LDPC codes in multicarrier underwater acoustic communications. We developed a design procedure to construct codes that match well with the underlying constellation, have excellent performance, and can be encoded in parallel and in linear time. We also demonstrated the use of LDPC codes to reduce the peak to average power ratio of OFDM transmissions. Extensive simulations were presented and field test results confirmed the code performance with real data. We plan to incorporate the proposed nonbinary LDPC codes into the DSP-based OFDM modem prototype presented in [47] for both performance improvement and PAPR reduction. Specifically, transmission modes in Table I can be adaptively used depending on the operating SNR.

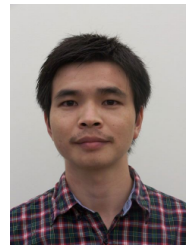
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Jie Huang was born in Jiangling, Hubei, P. R. China on January 20, 1981. He received the B.S. degree in 2001 and the Ph. D. degree in 2006, from the University of Science and Technology of China (USTC), Hefei, both in electrical engineering and information science. He has been a post-doctoral researcher with the Department of Electrical and Computer Engineering at the University of Connecticut (UConn), Storrs, since July 2007.

His general research interests lie in the areas of communications and signal processing, specifically error control coding theory and coded modulation system design. His recent focus is on signal processing, channel coding and network coding for underwater acoustic communications and underwater sensor networks. Mr. Huang has served as a reviewer for the IEEE Transactions on Communications, and the IEEE Journal on Selected Areas in Communications.



Shengli Zhou (M'03) received the B.S. degree in 1995 and the M.Sc. degree in 1998, from the University of Science and Technology of China (USTC), Hefei, both in electrical engineering and information science. He received his Ph.D. degree in electrical engineering from the University of Minnesota (UMN), Minneapolis, in 2002. He has been an assistant professor with the Department of Electrical and Computer Engineering at the University of Connecticut (UConn), Storrs, since 2003. He now holds a United Technologies Corporation

(UTC) Professorship in Engineering Innovation.

His general research interests lie in the areas of wireless communications and signal processing. His recent focus is on underwater acoustic communications and networking. Dr. Zhou has served as an Associate Editor for IEEE Transactions on Wireless Communications from Feb. 2005 to Jan. 2007, and is now an associate editor for IEEE Transactions on Signal Processing. He received the ONR Young Investigator award in 2007.



Peter Willett (F'03) received his B.A.Sc. (Engineering Science) from the University of Toronto in 1982, and his PhD degree from Princeton University in 1986. He has been a faculty member at the University of Connecticut ever since, and since 1998 has been a Professor. His primary areas of research have been statistical signal processing, detection, machine learning, data fusion and tracking. He has interests in and has published in the areas of change/abnormality detection, optical pattern recognition, communications and industrial/security condition monitoring.

He is editor-in-chief for IEEE Transactions on Aerospace and Electronic Systems, and until recently was associate editor for three active journals: IEEE Transactions on Aerospace and Electronic Systems (for Data Fusion and Target Tracking) and IEEE Transactions on Systems, Man, and Cybernetics, parts A and B. He is also associate editor for the IEEE AES Magazine, editor of the AES Magazines periodic Tutorial issues, associate editor for ISIFs electronic Journal of Advances in Information Fusion, and is a member of the editorial board of IEEE Signal Processing Magazine. He has been a member of the IEEE AESS Board of Governors since 2003. He was General Co-Chair (with Stefano Coraluppi) for the 2006 ISIF/IEEE Fusion Conference in Florence, Italy, Program Co-Chair (with Eugene Santos) for the 2003 IEEE Conference on Systems, Man, and Cybernetics in Washington DC, and Program Co-Chair (with Pramod Varshney) for the 1999 Fusion Conference in Sunnyvale.